Program**: Diploma in Renewable Energy**	
Course Code: **6298**	Course Title**: Simulation Lab**
Semester: **S6**	Credits**: 2.5**
Course Category: **Programme core**	
Periods per week: **4 (L: 0 T: 1 P: 3)**	Periods per semester: **60**

Course Objectives:

1. To train the students in Renewable Energy Sources and technologies.
2. To provide adequate inputs on a variety of issues in harnessing Renewable Energy.
3. To recognize the current and possible future role of Renewable energy sources.

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Principle and working of solar Thermal and Solar photovoltaic devices		Solar Energy Technology	III
Identify various components in balance of systems for PV installation		Renewable Energy Engineering -1	III

Course Outcomes:

On completion of the course student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Model a solar photovoltaic system using MATLAB - SIMULINK or any other software	15	Applying
CO2	Make use of the VI Characteristics to find the efficiency of a solar PV system.	15	Applying
CO3	Develop model of a micro–Wind Energy Generator	12	Applying
CO4	Develop the model of a hybrid power system and measure the performance	12	Applying
	Lab Test	6	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1			3				
CO2			3				
CO3			3				
CO4			3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Name of Experiment	Duration (Hours)	Cognitive Level
CO1	**Model a solar photovoltaic system using MATLAB - SIMULINK or any other software**		
M1.01	Modeling of PV cell-Effect Of Temperature Variation On Photovoltaic Array	9	Applying
M1.02	Effect of irradiation on a photovoltaic array	3	Applying
M1.03	Simulation of Buck & Boost converter	3	Applying
CO2	**Make use of the VI Characteristics to find the efficiency of a solar PV system.**		
M2.01	Design of solar PV boost converter using P&O MPPT technique	6	Applying
M2.02	Develop Shadowing effect & diode based solution in1kWp Solar PV system	3	Applying
M2.03	Model performance assessment of Grid connected and Standalone 1kWp Solar Power System.	6	Applying
	Lab Test 1	3	
C03	**Develop model of a micro–Wind Energy Generator**		
M3.01	Simulation study on Wind Energy Generator	6	Applying

M3.02	Simulation study on Intelligent Controllers for Hybrid Systems	3	Applying
M3.03	Experiment on Performance assessment of micro –Wind Energy Generator.	3	Applying
C04	**Develop the model of a hybrid power system and measure the performance**		
M4.01	Experiment on Performance Assessment of Hybrid (Solar-Wind) Power System.	6	Applying
M4.02	Simulation study on Hydel Power system	3	Applying
M4.03	Experiment on Performance Assessment of 100W fuel cell	3	Applying
	Lab test -2	3	Applying

Text / Reference

T/R	Book Title/Author
T1	Chetan Singh Solanki, Solar Photovoltaic – Fundamentals, Technologies and Applications, PHI Learning Private Limited.
T2	S. P. Sukhatme, Solar Energy – Principles of thermal collection and storage, second edition, Tata McGraw-Hil, New Delhi, 1996.
T3	Roger A. Messenger, Amir Abtahi, Photovoltaic Systems Engineering,4th Edition, CRC Press,2017
T4	Garg H. P. and Prakash J., Solar Energy – Fundamentals and Applications, Tata McGrawHill, 201
T5	Dictionary of RenewableResources-2ndEdition, Revised and Enlarged, Zoebelein, Hans,Wiley-VCH, 2001.
T6	.WindEnergyHandbook,TonyBurton,DavidSharpe,NickJenkins,ErvinBossanyi, JohnWiley&Sons;1st edition(2001)
T7	Fuel Cell System Explained James Larminie and Andrew Dicks, 2003, Pub WileyISBN:0-470-84857-X
T8	Energy conversion and storage scientific journals.

Online Resources

Sl.No	Website Link
1	https://www.nrel.gov/analysis/sam/help/htmlphp/index.html?libraries.htm.
2	National Renewable Energy Laboratory, USA. http://www.osti.gov/bridge

	Experiments with renewable energy-students guide- ISBN 1-928982-22-0, African Journal of Biotechnology, Vol. 9 (12), pp. 1719(2010).

Student Activity

Suggested Open-ended Experiments:

Students can do open ended experiments as a group of 3-5. There is no duplication in experiments in between groups. This is mainly for the purpose of continuous internal evaluation. Students should prepare a separate report on open ended experiment of their choice.

Examples.

1. Design of solar PV boost converter using incremental conductance technique (MPPT using incremental conductance (INC)with boost converter)
2. Simulation of Single Phase full converter using RLE loads
3. Simulation of a single phase AC Voltage Controller using RL loads.
4. Simulation of a single Phase Inverter with PWM control
5. Simulation of three phase PWM inverter
6. Simulation of Closed loop control of DC/DC converters